

Brain Tumor Classification with Bayesian Convolutional Neural Network: A Comparison of Variational Inference and Monte Carlo Dropout

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Received: 16 October 2020

Revised: 16 October 2020

Accepted: 16 October 2020

ABSTRACT – Deep learning models for brain tumor classification often produce overconfident predictions without indicating their reliability, which is problematic in medical diagnosis. This study applies Bayesian approximation methods, Monte Carlo (MC) Dropout and Variational Inference (VI), to a Convolutional Neural Network (CNN) for classifying brain Magnetic Resonance Imaging (MRI) scans into four classes: glioma, meningioma, pituitary, and no tumor. A dataset of 7,200 MRI images was divided into training, validation, and test sets (5,600/800/800) using stratified sampling and preprocessed through resizing, normalization, and data augmentation. Three models were evaluated: a baseline CNN, a Bayesian CNN with MC Dropout (BCNN-MC Dropout), and a Bayesian CNN with VI (BCNN-VI). Predictive uncertainty was quantified using predictive entropy and standard deviation across 30 stochastic forward passes. BCNN-VI achieved the highest test accuracy (94.63%), outperforming the baseline CNN (92.25%) and BCNN-MC Dropout (91.75%). Both Bayesian models assigned higher uncertainty to misclassified samples than to correctly classified ones, indicating that uncertainty can effectively signal potential prediction errors. These results demonstrate that VI can improve classification performance while providing interpretable confidence estimates, supporting its use for identifying ambiguous cases that may require expert review.

Keywords – Bayesian neural network, Posterior approximation, Uncertainty quantification, Medical image diagnosis

I. INTRODUCTION

Artificial Intelligence (AI) and Deep Learning (DL) have a significant contribution to many fields such as finance, engineering and maintenance, even in healthcare. However, because of its black box algorithm, either the traditional model or advanced, it makes it a struggle to communicate the uncertainty and difficult to interpret [1]. Because of this limitation, it can lead to overconfident predictions that make them unreliable for decision-making in medical contexts that have high consequences [2]. Bayesian models offer a solution to solve this issue because its framework assigns a probabilistic approach, so it develops the learning algorithm and makes it better than conventional DL in explaining the uncertainty [3][1].

Bayesian Neural Networks (BNNs) estimate the posterior distribution of parameters and facilitate the computation for both prediction and their corresponding uncertainty estimates. This capability is crucial in high-stakes environments where reliable estimations of predictive uncertainty are necessary for safe and accurate decision-making [2]. Especially in the medical field that requires not only the prediction outcome but also quantifying the uncertainty of the prediction, where decisions are often made under ambiguous conditions. BNN uses the prior distribution and data likelihood as the input elements to get the posterior distribution as the output. By integrating Bayesian inference into their network parameters, BNNs extend the capabilities of standard DL. This enables BNNs for both epistemic and aleatoric uncertainties, a crucial feature for safe clinical decision-making [4].

The posterior distribution is mathematically intractable in practical; to overcome this, approximation methods are employed to derive a tractable estimation of the model's weights. The two prevailing approximation methods are Variational Inference (VI) and Monte-Carlo Dropout (MC-Dropout). Variational inference (VI) is one of the approximation methods, it employed to approximate the weight posterior by minimizing the Negative Evidence Lower Bound (ELBO) during training [2]. The goal is to get an approximate distribution that closely estimates the posterior distribution by minimizing the Kullback-Leibler (KL) divergence which measures the amount of information loss while approximating a distribution between the true posterior and the variational distribution [1]. Conversely, MC-Dropout approximates the posterior distribution by randomly masking network weights using Bernoulli-distributed variables during both training and inference. Recent literature has demonstrated the efficacy of VI and MC-Dropout across various critical healthcare tasks, including lung cancer risk assessment [3], the detection of diabetic retinopathy [1], and the classification of Alzheimer's disease [2], underscoring its robustness in medical decision support systems.

Brain tumors (BT) present a compelling clinical challenge for Bayesian Neural Networks (BNNs) due to the high risks associated with their diagnosis. The lack of uncertainty quantification in standard DL limits its reliability in such scenarios. The fundamental benefit of this uncertainty quantification is its role as a safety mechanism, highlighting indeterminate cases that require expert human review and minimizing the risk of misdiagnosis. Therefore, this study proposes the application of BNNs to extend the diagnostic capabilities of standard DL. This research aims to apply BNN

techniques for BT image classification, with a primary focus on uncertainty quantification, and to compare VI and MC-Dropout in Bayesian Convolutional Neural Networks (BCNNs).

II. LITERATURE REVIEW

A. Image Pre-Processing

Image pre-processing is a critical stage in deep learning-based medical image classification, as it standardizes input data, improves computational efficiency, and enhances the robustness and generalization of the trained model, particularly in the analysis of medical images. **Resizing all images** to a uniform dimension is a widely adopted step in brain tumor Magnetic Resonance Imaging (MRI) classification studies, ensuring input consistency and compatibility with the fixed input shape required by Convolutional Neural Network (CNN) architectures, matching the input size required by the chosen model [5]. **Pixel intensity normalization**, typically performed by scaling pixel values to the $[0, 1]$ range, is similarly standard practice; this bounded range reduces numerical instability and improves the efficiency and stability of gradient-based optimization during training, making it easier for deep learning models to learn effective input-output mappings [5]. **Data augmentation**, including geometric and photometric transformations such as rotation, zooming, flipping, and brightness adjustment, is commonly applied exclusively to the training set to artificially diversify the training distribution, simulating real-world conditions and anatomical variations that improve the model's ability to generalize to unseen data, while mitigating overfitting that is especially prevalent in small or imbalanced medical imaging datasets, since deep learning models trained on limited data volumes are generally prone to overfitting [6]. Finally, **label encoding via one-hot representation** is a standard preparatory step for multi-class classification tasks trained with categorical cross-entropy loss, ensuring that class labels are represented in a format compatible with the softmax output layer of the network.

B. CNN (Convolutional Neural Network)

The proposed CNN architecture is inspired by the Visual Geometry Group (VGG) family of architectures (as shown in Fig 1), characterized by stacking pairs of small 3×3 convolutional layers before each downsampling stage. The VGG-based design was chosen due to its demonstrated effectiveness in prior brain tumor MRI classification studies; for instance, [7] employed a VGG16-based CNN for four-class brain tumor classification (glioma, meningioma, pituitary, and normal) and achieved a classification accuracy of 99.24% on a large combined dataset of 17,136 MRI images. Motivated by this demonstrated effectiveness, the present study adopts a VGG-style architecture built from scratch, rather than relying on pretrained ImageNet weights, in order to evaluate the Bayesian approximation methods (MC Dropout and VI) on a backbone architecture trained exclusively on the brain tumor MRI dataset itself. The network consists of four convolutional blocks, followed by a fully connected classification head. Each convolutional block contains two Conv2D layers with a 3×3 kernel and same padding (preserving spatial dimensions), with each convolution immediately followed by a BatchNormalization layer. After the second convolution-normalization pair, a MaxPooling2D layer with a 2×2 pool size is applied to reduce the spatial dimensions of the feature maps, followed by a Dropout layer for regularization. The number of filters doubles at each successive block (32, 64, 128, and 256) following the design principle in which shallow layers extract low-level features such as edges and textures using fewer filters, while deeper layers capture increasingly abstract and complex spatial patterns and therefore require a larger number of filters. ReLU activation is applied after every convolutional layer to introduce non-linearity. The dropout rate is also progressively increased across blocks (0.20, 0.25, 0.30, and 0.35, respectively), reflecting the higher overfitting risk associated with deeper layers that contain a larger number of parameters.

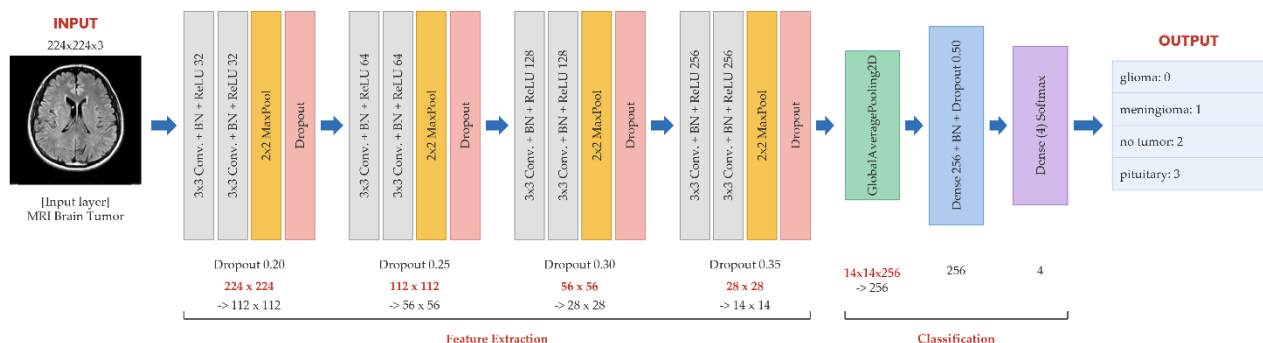


Figure 1 Illustration of the proposed CNN architecture

Following the four convolutional blocks, a GlobalAveragePooling2D layer is applied to condense each feature map into a single value, producing one feature vector per image. This approach is preferred over flattening, as it substantially reduces the number of parameters entering the dense layers and provides stronger regularization against overfitting. The

resulting vector is passed to a fully connected Dense layer of 256 units with ReLU activation, followed by a BatchNormalization layer and a Dropout layer with a rate of 0.5. The final output layer is a Dense layer with four units corresponding to the four tumor classes (glioma, meningioma, pituitary, and no tumor), using a softmax activation function to produce a normalized probability distribution over the classes. In total, this architecture consists of eight convolutional layers, four max-pooling layers, ten batch-normalization layers, five dropout layers, and two fully connected (dense) layers. To provide a clearer overview of the proposed network, Table 1 summarizes the complete layer-by-layer configuration of the custom CNN architecture, including the filter size, padding type, activation function, output shape, and number of trainable parameters at each layer.

Table 1 Details of the proposed CNN architecture.

Layer	Filter	Padding	Activation	Output Shape	No. of Parameters
Input	-	-	-	(224,224,3)	0
Conv2D	(32 × 3 × 3)	Same	ReLU	(224,224,32)	896
BatchNormalization	-	-	-	(224,224,32)	128
Conv2D	(32 × 3 × 3)	Same	ReLU	(224,224,32)	9,248
BatchNormalization	-	-	-	(224,224,32)	128
MaxPooling2D (2 × 2)	-	-	-	(112, 112,32)	0
Dropout (0.20)	-	-	-	(112, 112,32)	0
Conv2D	(64 × 3 × 3)	Same	ReLU	(112, 112,64)	18,496
BatchNormalization	-	-	-	(112, 112,64)	256
Conv2D	(64 × 3 × 3)	Same	ReLU	(112, 112,64)	36,928
BatchNormalization	-	-	-	(112, 112,64)	256
MaxPooling2D (2 × 2)	-	-	-	(56, 56,64)	0
Dropout (0.25)	-	-	-	(56, 56,64)	0
Conv2D	(128 × 3 × 3)	Same	ReLU	(56, 56,128)	73,856
BatchNormalization	-	-	-	(56, 56,128)	512
Conv2D	(128 × 3 × 3)	Same	ReLU	(56, 56,128)	147,584
BatchNormalization	-	-	-	(56, 56,128)	512
MaxPooling2D (2 × 2)	-	-	-	(28, 28,128)	0
Dropout (0.30)	-	-	-	(28, 28,128)	0
Conv2D	(256 × 3 × 3)	Same	ReLU	(28, 28,256)	295,168
BatchNormalization	-	-	-	(28, 28,256)	1,024
Conv2D	(256 × 3 × 3)	Same	ReLU	(28, 28,256)	590,080
BatchNormalization	-	-	-	(28, 28,256)	1,024
MaxPooling2D (2 × 2)	-	-	-	(14, 14,256)	0
Dropout (0.35)	-	-	-	(14, 14,256)	0
GlobalAveragePooling2D	-	-	-	(256)	0
Dense	-	-	-	(256)	65,792
BatchNormalization	-	-	-	(256)	1,024
Dropout (0.5)	-	-	-	(256)	0
Dense	-	-	Softmax	(4)	1,028
Total					1,243,940

C. Uncertainty Weight in Bayesian CNN

In the Bayesian CNN framework, model parameters are treated as random variables rather than deterministic values, illustrated in Figure 2 [8]. Specifically, each weight in the CNN network is assumed to follow a probability distribution, typically a Gaussian distribution:

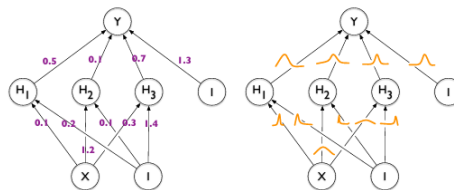


Figure 2 Left: each weight has a fixed value, as provided by classical backpropagation. Right: each weight is assigned a distribution, as provided by Bayesian method.

$$w \sim \mathcal{N}(\mu, \sigma^2) \quad (1)$$

where μ represents the mean and σ^2 denotes the variance of the weight distribution. This formulation theoretically enables the model to capture epistemic uncertainty, which arises from limited data and uncertainty in model parameters [9]. Given a training dataset and supervised learning environment, $D = \{\mathbf{x}_n, \mathbf{y}_n\}_{n=1}^N$, where N represents the size of the

dataset, $\mathbf{x}_n$ represents a vector of input features, and $\mathbf{y}_n$ denotes the corresponding one-hot encoded label vector over the four tumor classes (glioma, meningioma, no tumor, pituitary). The objective of Bayesian method is to estimate the posterior uncertainty on the model parameters $p(\mathbf{w}|D)$, by assuming a distribution of prior on neural network parameters $p(\mathbf{w})$. Model ($\hat{y}$) over a testing data input ($\mathbf{x}^*$) can be computed by:

$$p(\hat{y}|\mathbf{x}^*, D) = \int p(\hat{y}|\mathbf{x}^*, \mathbf{w})p(\mathbf{w}|D)d\mathbf{w} \quad (2)$$

where $p(\mathbf{w}|D)$ is the posterior distribution of weights given data D . However, this integral is analytically intractable for deep neural networks, because $p(\mathbf{w}|D)$ is intractable, motivating the use of approximate inference methods [10]. By representing weights probabilistically, BCNN not only produces point predictions but also quantifies predictive uncertainty.

D. Variational Inference as Bayesian Approximation

VI is a method used to approximate the intractable posterior distribution $p(\mathbf{w}|D)$ by introducing a simpler variational distribution $q(\mathbf{w})$ [11]. The objective is to find $q(\mathbf{w})$ that is closest to the true posterior ($p(\mathbf{w}|D)$). A measure for this distance is the Kullback-Leibler (KL) divergence, defined as:

$$D_{KL}(q(\mathbf{w})||p(\mathbf{w}|D)) = \int q(\mathbf{w})\log \frac{q(\mathbf{w})}{p(\mathbf{w}|D)}d\mathbf{w} \quad (3)$$

since direct minimization is intractable, the minimization of the KL divergence is equivalent to the reduction of the negative Evidence Lower Bound [11], which serves as a lower bound to the log marginal likelihood:

$$\mathcal{L}(w_i) = -\mathbb{E}_{q(\mathbf{w})}[\log p(D|\mathbf{w}_i)] + KL(q(\mathbf{w}_i)||p(\mathbf{w}_i)) \quad (4)$$

where $\mathcal{L}(w_i)$ denotes the loss function (negative ELBO) to be minimized for weight w_i , $\mathbb{E}_{q(\mathbf{w}_i)}[\cdot]$ denotes the expectation taken with respect to the variational posterior distribution $q(\mathbf{w}_i)$, $p(D|\mathbf{w}_i)$ denotes the likelihood of the observed dataset D given weight w_i , and $KL(q(\mathbf{w}_i)||p(\mathbf{w}_i))$ denotes the Kullback-Leibler divergence between the variational posterior $q(\mathbf{w}_i)$ and the prior $p(\mathbf{w}_i)$, as defined in Eq. (3). The inference process begins by inputting the observed data y_0 , defining the three main components of the Bayesian framework: the prior distribution $p(\mathbf{w})$, the likelihood $p(y_0|\mathbf{w})$, and the variational distribution $q(\mathbf{w}|\lambda)$, which serves as an approximation to the true posterior. The variational parameters $\lambda = \{\lambda_1, \lambda_2, \dots, \lambda_n\}$ are then initialized, corresponding to the neural network parameters $\mathbf{w} = \{w_1, w_2, \dots, w_n\}$. The optimization proceeds iteratively, where for each parameter λ_i , the gradient of the ELBO with respect to λ_i is computed, and the parameter is updated accordingly. After updating, the ELBO is recalculated using the new λ_i based on the following formulation (4) above. The total ELBO is obtained by summing over all parameters, $\text{ELBO} = \sum_{i=1}^N \mathcal{L}(\lambda_i)$. This iterative process continues by increasing the iteration index and repeating the update steps until convergence is achieved, meaning the ELBO does not change significantly across several iterations. VI in CNN models is highly complex because it requires defining and optimizing a probability distribution over every network parameter, including weights and biases across multiple convolutional layers. This involves computing ELBO and performing stochastic sampling during both training and inference, which significantly increases computational cost and implementation difficulty.

The prior distribution expresses the assumed distribution over each network parameter before observing the data. In this study, the prior for each weight w_i is defined as a fixed standard normal distribution, independent across all parameters and not updated during training. This prior can be written as:

$$p(w_i) \sim \mathcal{N}(0,1) \quad (5)$$

in the variational inference, the posterior distribution is approximated with the corresponding variational distribution. In this study, the variational distribution of each parameter in CNN follows a normal distribution. The variational parameters λ_i for normal distribution denote the mean and variance. By updating the means and variance of the variational distribution (which explains why all combination λ_i is being tried), variational distribution will be closer to the posterior distribution. The variational posterior for each weight, along with its initialization scheme, can therefore be written as:

$$q(w_i) \sim \mathcal{N}(\mu_i, \sigma_i^2) \quad (6)$$

$$\mu_i \sim \mathcal{N}(\alpha_i, \gamma_i^2), \sigma_i = \text{softplus}(\rho_i), \rho_i \sim \mathcal{N}(-3, 0.1^2)$$

where $\alpha_i = 0$ and $\gamma_i = 0.1$ denote the default initialization hyperparameters for the posterior mean and scale, respectively, following TensorFlow Probability's mean-field normal initialization scheme (default_mean_field_normal_fn). These initial values determine the starting point of the variational distribution for each weight prior to training; as training proceeds, μ_i and σ_i are iteratively updated via gradient-based optimization of the ELBO objective in Eq. (4), allowing the variational posterior $q(w_i)$ to progressively approach the true posterior distribution $p(\mathbf{w}|D)$.

E. Monte Carlo Dropout as Bayesian Approximation

MC Dropout, originally designed as a regularization technique, has been reinterpreted as a method for approximate Bayesian inference, enabling the CNN to quantify predictive uncertainty [12]. This is achieved through a regularization mechanism that randomly deactivates neurons at each training step, where each weight matrix is randomly masked using Bernoulli-distributed variables:

$$w_i = M_i \cdot \text{diag}(z_i), \quad z_i \sim \text{Bernoulli}(p_i) \quad (7)$$

where M_i denotes the base weight matrix learned through standard backpropagation, and z_i denotes a binary mask vector

sampled from a Bernoulli distribution with dropout probability p_i , applied element-wise to deactivate individual neurons. This formulation implies that dropout training is equivalent to minimizing the Kullback-Leibler (KL) divergence between the approximate distribution $q(\mathbf{w})$ and the true posterior $p(\mathbf{w} | D)$ [12]. At the inference stage, instead of using deterministic weights, MC Dropout performs multiple stochastic forward passes by sampling from $q(\mathbf{w})$. The predictive distribution is approximated as:

$$q(\mathbf{y}^* | \mathbf{x}^*) \propto \int p(\mathbf{y}^* | \mathbf{x}^*, \mathbf{w}) q(\mathbf{w}) d\mathbf{w} \quad (8)$$

Since evaluating this integral is computationally expensive, Monte Carlo integration is used to approximate it. By performing T stochastic forward passes with dropout, MC Dropout generates a set of prediction $\{y_1^*, y_2^*, \dots, y_T^*\}$. The empirical mean and variance of these predictions serve as approximations for the predictive mean and uncertainty respectively.

$$\bar{y}^* = \frac{1}{T} \sum_{t=1}^T y_t^*; \quad \sigma_{y^*}^2 = \frac{1}{T} \sum_{t=1}^T (y_t^* - \bar{y}^*)^2 \quad (9)$$

F. CNN Architecture for BCNN-VI

BCNN-VI model adopts the same VGG-style backbone structure as the baseline CNN described in Table 1, including the number of convolutional blocks, filter progression ($32 \rightarrow 64 \rightarrow 128 \rightarrow 256$), padding scheme, and dropout rates. However, each standard Conv2D and Dense layer is replaced with its probabilistic counterpart, Conv2DFlipout and DenseFlipout, in which the kernel weights are treated as learnable distributions rather than fixed point estimates. The complete layer-by-layer configuration of the BCNN-VI model is presented in Table 2.

Table 2 Details of the proposed BCNN-VI architecture.

Layer	Filter	Padding	Activation	Output Shape	No. of Parameters
Input	-	-	-	(224,224,3)	0
Conv2DFlipout	(32 × 3 × 3)	Same	ReLU	(224,224,32)	1,760
BatchNormalization	-	-	-	(224,224,32)	128
Conv2DFlipout	(32 × 3 × 3)	Same	ReLU	(224,224,32)	18,464
BatchNormalization	-	-	-	(224,224,32)	128
MaxPooling2D (2 × 2)	-	-	-	(112, 112,32)	0
Dropout (0.20)	-	-	-	(112, 112,32)	0
Conv2DFlipout	(64 × 3 × 3)	Same	ReLU	(112, 112,64)	36,928
BatchNormalization	-	-	-	(112, 112,64)	256
Conv2DFlipout	(64 × 3 × 3)	Same	ReLU	(112, 112,64)	73,792
BatchNormalization	-	-	-	(112, 112,64)	256
MaxPooling2D (2 × 2)	-	-	-	(56, 56,64)	0
Dropout (0.25)	-	-	-	(56, 56,64)	0
Conv2DFlipout	(128 × 3 × 3)	Same	ReLU	(56, 56,128)	147,584
BatchNormalization	-	-	-	(56, 56,128)	512
Conv2DFlipout	(128 × 3 × 3)	Same	ReLU	(56, 56,128)	295,040
BatchNormalization	-	-	-	(56, 56,128)	512
MaxPooling2D (2 × 2)	-	-	-	(28, 28,128)	0
Dropout (0.30)	-	-	-	(28, 28,128)	0
Conv2DFlipout	(256 × 3 × 3)	Same	ReLU	(28, 28,256)	590,080
BatchNormalization	-	-	-	(28, 28,256)	1,024
Conv2DFlipout	(256 × 3 × 3)	Same	ReLU	(28, 28,256)	1,179,904
BatchNormalization	-	-	-	(28, 28,256)	1,024
MaxPooling2D (2 × 2)	-	-	-	(14, 14,256)	0
Dropout (0.35)	-	-	-	(14, 14,256)	0
GlobalAveragePooling2D	-	-	-	(256)	0
DenseFlipout	-	-	-	(256)	131,328
BatchNormalization	-	-	-	(256)	1,024
Dropout (0.5)	-	-	-	(256)	0
DenseFlipout	-	-	Linear	(4)	2,052
			(logits)		
Softmax	-	-	Softmax	(4)	0
Total					2,481,796

G. Model Uncertainties

Uncertainty quantification is a crucial aspect of predictive modelling, which measures how certain the model is. In Bayesian modelling, two fundamental types of uncertainties, namely Aleatoric and Epistemic, are often considered [4]. We apply VI approximation and MC-Dropout in the CNN model to estimate its uncertainty. Final classifier is determined based on Eq. (10), where the category is assigned based on the highest mean prediction probability. The predictive

standard deviation and predictive entropy across T ($T = 30$) stochastic forward passes are used to estimate the uncertainty of the Bayesian CNN. The formula for predictive entropy is as follows:

$$H_p(\hat{y}|\mathbf{x}^*) = - \sum_c \bar{y}_c \log \bar{y}_c \quad (10)$$

H. Evaluation Metrics

Once the developed models have been implemented and trained, the subsequent step involves applying evaluation metrics to assess their performance. In this work, several evaluation methods were employed. First, comparison evaluation metrics, including accuracy, recall, precision, and F1-score were utilized to compare the predicted and actual values. Secondly, a normalized confusion matrix was presented, which provides a comprehensive insight of the model performance. Thirdly, the evaluation of the model's classification task was analyzed utilizing the Receiver Operating Characteristics (ROC) curve along with the area under the ROC curve (AUC). The metrics used are defined as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (11)$$

$$Precision = \frac{TP}{TP + FP} \quad (12)$$

$$Recall = \frac{TP}{TP + FN} \quad (13)$$

$$F1 - Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (14)$$

Finally, we calculated the entropy to measure uncertainty in data prediction for Bayesian CNN model. Entropy is a measure of disorder or randomness in the data, and it is used to quantify the uncertainty associated with the predicted values. The entropy value ranges from 0 to $\ln(C)$, where C denotes the number of classes, and represents the level of uncertainty and disorder in the model's predictions. In this study, with $C = 4$ tumor classes, the predictive entropy ranges from 0 to $\ln(4) \approx 1.386$. The primary goal of this model was to reduce the uncertainty while maintaining as low entropy as possible.

III. METHODOLOGY

The dataset used in this study is the Brain Tumor MRI Dataset obtained from a curated combination of publicly available datasets, such as the figshare brain MRI dataset, SARTAJ dataset, and Br35H dataset, which contains 7,200 labeled magnetic resonance imaging (MRI) scans of the human brain. These brain tumor images were distributed across four classes as follows: 1800 images with Glioma (0), 1800 images with Meningioma (1), 1800 images with No Tumor (2), and 1800 images with Pituitary (3), with images sizes ranging from 150×167 to 1375×1446 pixels. To address these challenges, this study employs a Convolutional Neural Network (CNN) integrated with Bayesian inference (Bayesian-CNN). Specifically, posterior approximation is performed using Monte Carlo (MC) Dropout and Variational Inference (VI). This integration aims to quantify predictive uncertainty in the classification results, enabling the model not only to make predictions but also to assess its confidence, thereby providing more reliable and informative outputs for decision-making. In addition, all data processing and model development in this study were conducted using the Python programming language, supported by libraries such as TensorFlow, TensorFlow Probability, and OpenCV, and implemented within the Kaggle environment. The following are the stages of analysis in this study.

1. Collect the brain tumor MRI dataset (Masoud Nickparvar dataset), consisting of images labeled into four classes: glioma, meningioma, pituitary, and no tumor.
2. Conduct exploratory data analysis
3. Split the dataset into training, validation, and test sets using stratified sampling to preserve class proportions across all splits. The dataset, consisting of 7,200 MRI images in total, was divided into 5,600 images (77.78%) for training, 800 images (11.11%) for validation, and 800 images (11.11%) for testing — approximately a 7:1:1 ratio. The validation and test sets were derived by splitting the original test partition equally (50:50) using stratified sampling, ensuring each of the four tumor classes (glioma, meningioma, pituitary, no tumor) was equally represented with 200 images per class in both the validation and test sets.
4. Conduct data pre-processing, including resize all images to a uniform input dimension of 224×224 pixels, normalize pixel intensity values to range $[0, 1]$ through rescaling, apply data augmentation (rotation ranging from $-10^\circ - 10^\circ$, zoom ($\pm 10\%$), horizontal flip (50% probability), and brightness adjustment ($\pm 10\%$)) to the training set only (reduce overfitting), and finally encode class labels into one-hot vectors to match the categorical cross-entropy loss function.
5. Design the baseline CNN architecture (VGG-style), consisting of four convolutional blocks with paired 3×3 convolutions, batch normalization, max-pooling, and dropout, followed by a global average pooling layer and a fully connected classification head with softmax activation.
6. Design the first Bayesian CNN model using the Monte Carlo (MC) Dropout approximation.
7. Design the second Bayesian CNN model using Variational Inference (VI), in which convolutional and dense layers are replaced with Flipout layers (Convolution2DFlipout, DenseFlipout) that treat each weight as a learnable variational distribution.

8. Train the baseline CNN, BCNN-MC Dropout, and BCNN-VI model using the Adam optimizer, batch size = 32, with an initial learning rate of 0.001 and categorical cross-entropy loss, for a maximum of 100 epochs. Two callbacks were employed to regulate training: (1) Early Stopping, which monitors the validation loss and halts training if no improvement is observed for 8 consecutive epochs, restoring the model weights from the epoch with the lowest validation loss; and (2) Learning Rate Reduction on Plateau, which monitors the validation loss and reduces the learning rate by a factor of 0.5 if no improvement is observed for 4 consecutive epochs, down to a minimum learning rate of 1×10^{-6} . Since the patience for learning rate reduction (4 epochs) is shorter than that for early stopping (8 epochs), the learning rate is reduced at least once before training is terminated, allowing the model an opportunity to recover from a plateau before stopping altogether.
9. Perform inference under the BCNN-MC Dropout and BCNN-VI model using $T = 30$ stochastic forward passes (Flipout sampling), then compute the mean prediction and predictive standard deviation across passes.
10. Evaluate the baseline CNN, BCNN-MC Dropout, and BCNN-VI model on the test set using accuracy, classification report (precision, recall, F1-score), confusion matrix, and ROC-AUC curves (for BCNN, based on the mean prediction across the T passes).
11. Quantify predictive uncertainty for the BCNN-MC Dropout and BCNN-VI model by computing the predictive entropy and predictive standard deviation for each test sample.
12. Compare the overall performance and predictive uncertainty between the baseline CNN, BCNN-MC Dropout, and BCNN-VI models using descriptive statistics and visualization (e.g., boxplots of entropy and standard deviation).

IV. RESULTS AND DISCUSSIONS

A. Exploratory Data Analysis (EDA)

The class labels were derived directly from the original folder structure of the Masoud Nickparvar brain tumor MRI dataset, in which each image is pre-sorted into one of four category folders (glioma, meningioma, pituitary, and no tumor) without requiring additional manual annotation. These class labels were subsequently extracted programmatically and assigned to each image as its corresponding ground-truth label, as summarized in Table 3 and illustrated in Figure 3. During the pre-processing stage, these categorical labels were further encoded into one-hot vectors to match the categorical cross-entropy loss function used in model training.

Table 3 Description of each data class.

Class	Description
Glioma (Label = 0)	Gliomas are the most prevalent type of primary brain tumor, originating from glial cells that support neurons in the brain [13].
Meningioma (Label = 1)	Meningiomas are the most common primary intracranial tumor, arising from the meninges (the membranes surrounding the brain and spinal cord) [13].
No Tumor (Label = 2)	MRI scans showing normal brain tissue with no evidence of glioma, meningioma, or pituitary tumor, used as the negative/control class in the classification task [13].
Pituitary (Label = 3)	Pituitary tumors are abnormal growths that form in the pituitary gland. They are very common, but fortunately, most are invariably benign [13].

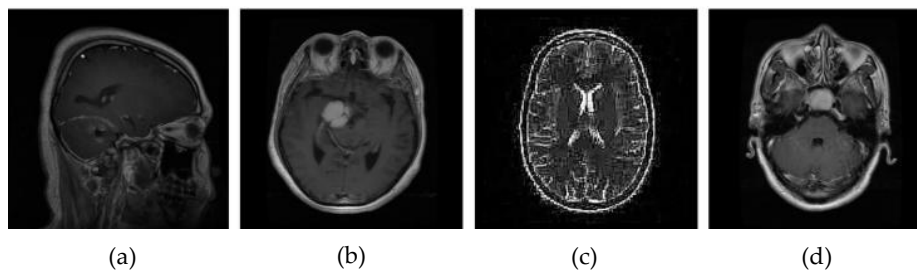


Figure 3 Sample image of (a) Glioma; (b) Meningioma; (c) No Tumor; (d) Pituitary.

B. Images Pre-Processing

The raw dataset was processed into three subsets: training data, validation data, and test data. The original Training folder was used entirely as the training set, while the original Testing folder was further split equally (50:50) into validation and test sets using stratified random sampling to preserve class proportions. The resulting data partition is presented in Table 4.

Table 4 Data splitting mechanism.

Subset	Number of Images	Class and Count
Training	5600	Glioma (1400), Meningioma (1400), Notumor (1400), and Pituitary (1400)
Validation	800	Glioma (200), Meningioma (200), Notumor (200), and Pituitary (200)
Test	800	Glioma (200), Meningioma (200), Notumor (200), and Pituitary (200)

Following the data partitioning, three pre-processing techniques were applied: resizing, rescaling, and data augmentation. The original images varied in resolution, ranging from 150×167 to 1375×1446 pixels. To ensure a uniform input dimension compatible with the CNN architecture, all images were first resized to 224×224 pixels, consistent with the standard input size adopted in VGG-based architectures [14]. Following resizing, rescaling was applied uniformly to all three subsets (training, validation, and test) by dividing every pixel intensity value by 255, transforming the original pixel range of $[0, 255]$ into a normalized range of approximately $[0, 1]$. This normalization step reduces large variations in pixel intensity that can otherwise impair learning, resulting in more stable gradients and faster convergence during training [15]. In contrast to resizing and rescaling, data augmentation was applied exclusively to the training set to artificially increase the diversity of training samples and reduce the risk of overfitting. Four augmentation techniques were employed: random rotation ($\pm 10^\circ$), random zoom ($\pm 10\%$), random horizontal flip (50% probability), and random brightness adjustment (90–110% of the original brightness). Figure 4 illustrates an example of a single glioma image subjected to multiple rounds of random augmentation, where each output exhibits slight variations in orientation, scale, and brightness while preserving the overall anatomical structure of the brain.

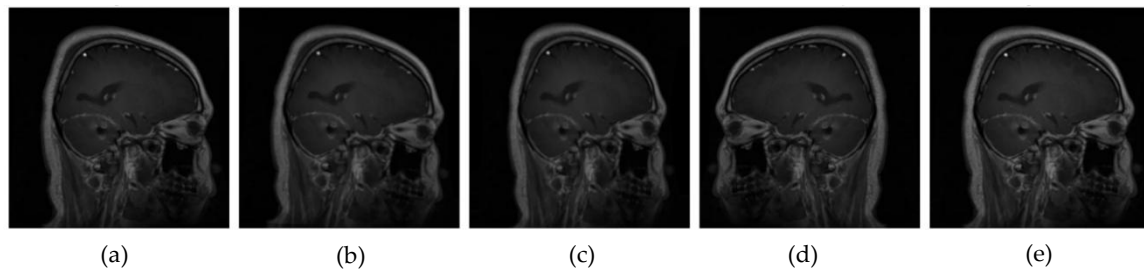


Figure 4 Illustration of data augmentation techniques applied to a glioma MRI image: (a) original image, (b) random rotation, (c) random zoom, (d) horizontal flip, and (e) brightness adjustment.

C. Training Process

After the pre-processing phase, the resulting dataset was used to train the baseline CNN model for a maximum of 100 epochs. The Adam optimization algorithm was employed during training with an initial learning rate of 0.001. To assess the model's performance during training, categorical cross-entropy was used as the loss function. To classify the input images into four classes, the output layer of the CNN model was composed of a fully connected (Dense) layer with a Softmax activation function. To prevent overfitting and ensure efficient training, two callbacks were applied, following the optimization strategy proposed by [16]: Early Stopping, which halts training if the validation loss does not improve for 8 consecutive epochs and restores the best-performing weights, and Reduce Learning Rate on Plateau, which reduces the learning rate by a factor of 0.5 if the validation loss does not improve for 4 consecutive epochs, down to a minimum learning rate of 1×10^{-6} . To train the BCNN models, we utilized the same VGG-style backbone architecture developed for the baseline CNN, consisting of four convolutional blocks with progressively increasing filter sizes (32, 64, 128, 256).

For the **MC-Dropout model**, dropout was applied after each convolutional block with progressively increasing rates (0.20, 0.25, 0.30, and 0.35 for the first through fourth blocks, respectively) and a rate of 0.5 in the dense layer, with dropout kept active during both training and inference (training=True) to enable stochastic forward passes. The model was trained using the Adam optimizer with a learning rate of 0.001, categorical cross-entropy as the loss function, and the same Early Stopping (patience of 8 epochs) and Reduce Learning Rate on Plateau (patience of 4 epochs, factor of 0.5, minimum learning rate of 1×10^{-6}) configuration used for the baseline CNN, for a maximum of 100 epochs.

For the **VI model**, we used TensorFlow Probability and implemented Convolution2DFlipout and DenseFlipout layers in place of standard convolutional and dense layers, allowing each kernel weight to be treated as a learnable distribution rather than a fixed-point estimate. The kernel posterior was initialized using TensorFlow Probability's default mean-field normal function, while the kernel prior was set to the default multivariate standard normal distribution. The loss function combined the categorical cross-entropy (negative log-likelihood) with the KL divergence between the posterior and prior distributions, scaled by a KL weight of 1×10^{-5} and normalized by the number of training samples, in accordance with the ELBO objective. The VI model was trained using the Adam optimizer with a learning rate of 0.001, for a maximum of 100 epochs, using a longer patience configuration than the previous two models (16 epochs for Early Stopping and 8 epochs

for Reduce Learning Rate on Plateau) to accommodate the noisier convergence behavior introduced by stochastic weight sampling during training. To compare the training behavior of the three models, Table 5 summarizes the final training accuracy, the number of epochs required for convergence (i.e., the epoch at which Early Stopping restored the best weights), and the total training runtime for the baseline CNN, BCNN-MC Dropout, and BCNN-VI models.

Table 5 Train performance comparison of CNN and Bayesian CNN models.

Model	Train Accuracy	Convergence Epoch	Runtime
CNN	0.9836	24	0 h 43 m 19 s
BCNN-MC Dropout	0.9511	22	0 h 39 m 34 s
BCNN-VI	0.9677	56	1 h 48 m 29 s

As shown in Table 5, the baseline CNN achieved the highest training accuracy (0.9836) and converged the fastest in terms of wall-clock time, requiring approximately 43 minutes to reach its best weights at epoch 24. The BCNN-MC Dropout model converged in a comparable number of epochs (epoch 22) and required the shortest overall training time (approximately 40 minutes), despite sharing the exact same architecture and parameter count as the baseline CNN, suggesting that the additional stochasticity introduced by active dropout during training did not meaningfully hinder convergence speed. In contrast, the BCNN-VI model required substantially more epochs to converge (epoch 56) and nearly 2.5 times longer training time (approximately 108 minutes) compared to the other two models, despite achieving comparable training accuracy (0.9807). This slower and more computationally expensive convergence is consistent with the nature of variational inference, where each forward pass involves sampling stochastic weights from learned posterior distributions rather than using fixed point estimates, in addition to the model needing to jointly optimize both the categorical cross-entropy and the KL-divergence term as part of the ELBO objective [8], resulting in a noisier and more complex optimization landscape than that of the deterministic CNN or the dropout-based approximation.

D. Model Evaluation (Test Dataset)

As shown in Table 6, the BCNN-VI model obtains the highest metric values among all evaluated models, achieving an accuracy of 0.9463, precision of 0.9525, recall of 0.9425, and F1-score of 0.9450, followed by the standard CNN with an accuracy of 0.9225 and F1-score of 0.9211, while BCNN-MC Dropout yields the lowest performance with an accuracy of 0.9175 and F1-score of 0.9164. The superior performance of BCNN-VI suggests that variational inference is able to effectively capture uncertainty in the network weights while simultaneously learning more robust feature representations, which can improve the model's generalization ability and reduce overfitting. In contrast, although MC Dropout also introduces Bayesian uncertainty estimation, the stochastic masking process during inference may lead to less stable feature utilization, resulting in a slight decrease in predictive performance. These findings indicate that incorporating Bayesian principles does not necessarily compromise classification accuracy; instead, when implemented through variational inference, Bayesian CNNs can provide both uncertainty awareness and improved predictive performance compared with conventional deterministic CNNs, consistent with findings reported in prior studies [17].

Table 6 Performance comparison of CNN and Bayesian CNN models.

Model	Accuracy	Precision	Recall	F1-score
CNN	0.9225	0.9248	0.9225	0.9211
BCNN-MC Dropout	0.9175	0.9177	0.9175	0.9164
BCNN-VI	0.9463*	0.9525*	0.9425*	0.9450*

As shown in Figure 5, the models exhibit distinct learning behaviors. The CNN model (a) achieves the highest training accuracy, rapidly reaching near-perfect values; however, it displays a significant gap between training and validation curves, indicating signs of overfitting. In contrast, the Bayesian CNN variants demonstrate superior generalization capabilities. The BCNN-MC Dropout model (b) shows a more balanced learning trajectory, successfully mitigating the overfitting observed in the standard CNN by narrowing the gap between training and validation accuracy, despite some observed fluctuations, consistent with the well-established role of dropout as a regularization mechanism that reduces co-adaptation among neurons and improves generalization [18]. Meanwhile, the BCNN-VI model (c) exhibits the most stable training behavior among the three. Although it reaches a slightly lower peak accuracy compared to the standard CNN, the training and validation curves remain remarkably close throughout the epochs. This consistency suggests that the BCNN-VI model is more robust and reliable, providing better generalization performance on unseen data by effectively regularizing the learning process, a finding consistent with prior work demonstrating that the KL-divergence term in the variational inference objective acts as a natural regularizer, leading to more stable training dynamics compared to deterministic networks [8].

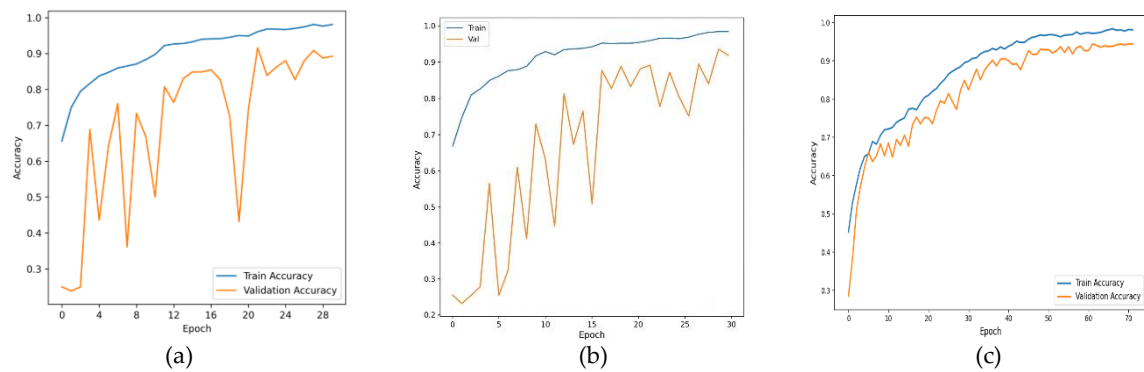


Figure 5 Train and validation accuracy curves of (a) CNN; (b) BCNN-MC Dropout; (c) BCNN-VI.

As shown in Figure 6, all models exhibit a decreasing training loss throughout the training process, indicating successful learning of underlying patterns from the data. The CNN model demonstrates a rapid reduction in training loss, while its validation loss decreases overall but remains highly fluctuating, suggesting that the model may be more sensitive to variations in the validation data despite achieving good predictive performance. A similar pattern is observed in the BCNN-MC Dropout model, where the validation loss exhibits even larger fluctuations across epochs, reflecting the stochastic nature introduced by Monte Carlo sampling and dropout-based uncertainty estimation. In contrast, the BCNN-VI model shows the smoothest and most stable convergence behavior, with both training and validation losses decreasing consistently over a longer training period and maintaining a relatively small gap between them. Although its loss values remain higher than the final training loss of the other models, the stable optimization trajectory suggests better regularization and generalization capability. This behavior is consistent with the superior classification performance achieved by BCNN-VI in Table 6, indicating that variational inference effectively balances uncertainty modeling and predictive accuracy while reducing the risk of overfitting.

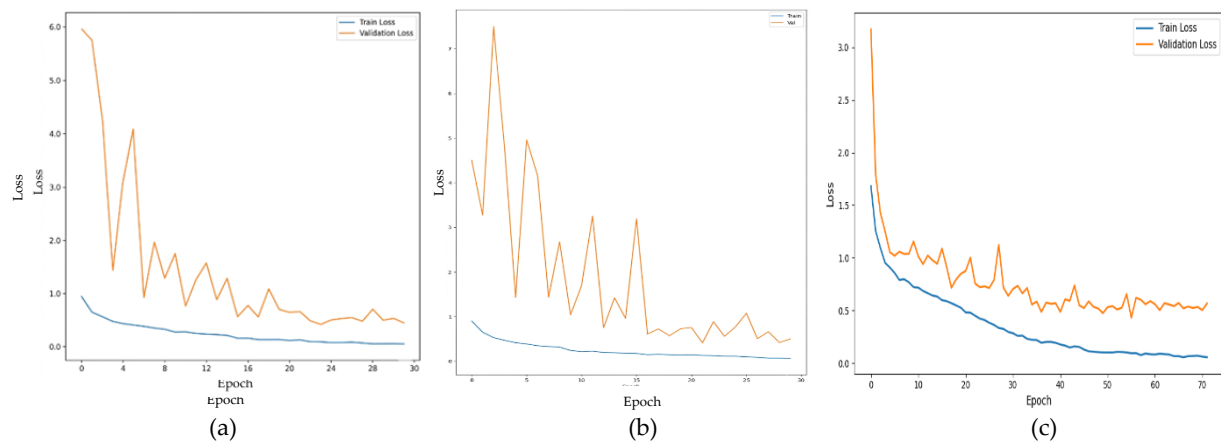


Figure 6 Train and validation loss curves of (a) CNN; (b) BCNN-MC Dropout; (c) BCNN-VI.

All three models exhibit a strong diagonal structure in their confusion matrices, indicating that the majority of samples are correctly classified, shown in Figure 7. Among them, the BCNN-VI model demonstrates the strongest diagonal dominance, particularly for the notumor and pituitary classes, with 200 and 197 correctly classified samples, respectively, and fewer overall misclassifications compared to the other models. The CNN model also achieves high classification performance, although a notable number of glioma samples are misclassified as notumor (22 cases) and meningioma (12 cases). Similarly, the BCNN-MC Dropout model maintains a clear diagonal pattern but exhibits the highest level of confusion among the three models, especially for the glioma class, where misclassifications to meningioma (16 cases) and notumor (15 cases) occur more frequently. Across all models, the most common source of error involves distinguishing glioma from meningioma and notumor, suggesting that these classes share overlapping imaging characteristics that make them more difficult to separate, consistent with prior findings in brain tumor MRI classification [19]. In contrast, the notumor and pituitary classes are consistently recognized with very high accuracy, indicating that their visual features are more distinctive. Overall, the confusion matrices support the quantitative results in Table 6, where BCNN-VI achieves

the best classification performance by producing the fewest misclassifications and the clearest class separation among the evaluated models.

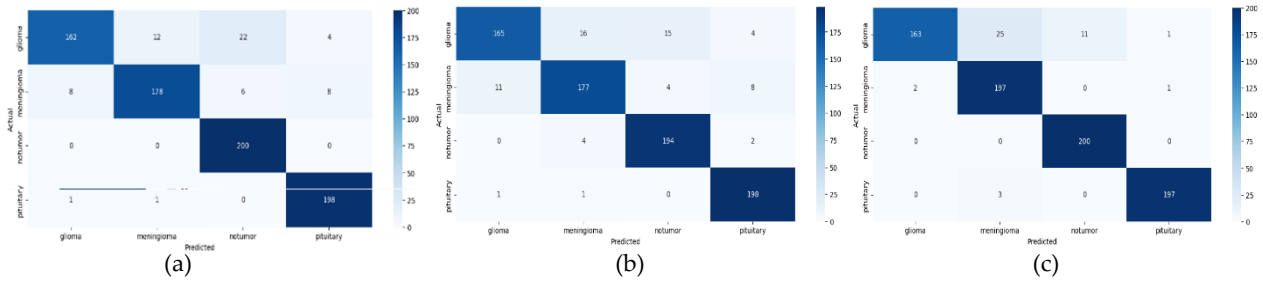


Figure 7 Confusion matrices of (a) CNN; (b) BCNN-MC Dropout; (c) BCNN-VI.

As shown in Figure 8, all three models exhibit excellent discriminative capability, as reflected by ROC curves that are concentrated near the upper-left corner and AUC values approaching 1.0 for most classes. The CNN model achieves strong performance, particularly for the pituitary (AUC = 0.9993), notumor (AUC = 0.9971), and meningioma (AUC = 0.9885) classes, while glioma shows the lowest AUC (0.9528), indicating greater classification difficulty. The BCNN-MC Dropout model demonstrates a similar trend, with near-perfect separability for pituitary and notumor, but slightly lower performance for glioma and meningioma, consistent with its lower overall classification metrics. In contrast, the BCNN-VI model produces the most favorable ROC curves overall, with all classes achieving very high AUC values and the glioma class showing a noticeable improvement compared to the other models. This result suggests that variational inference enhances the model's ability to distinguish between challenging classes by learning more robust probabilistic representations of the data. Across all models, notumor and pituitary remain the easiest classes to separate, as evidenced by their near-perfect AUC values, whereas glioma consistently presents the greatest challenge due to its higher overlap with other tumor categories [19]. Overall, the ROC analysis supports the findings from the confusion matrices and classification metrics, confirming that BCNN-VI provides the strongest class separability and overall discrimination performance among the evaluated models.

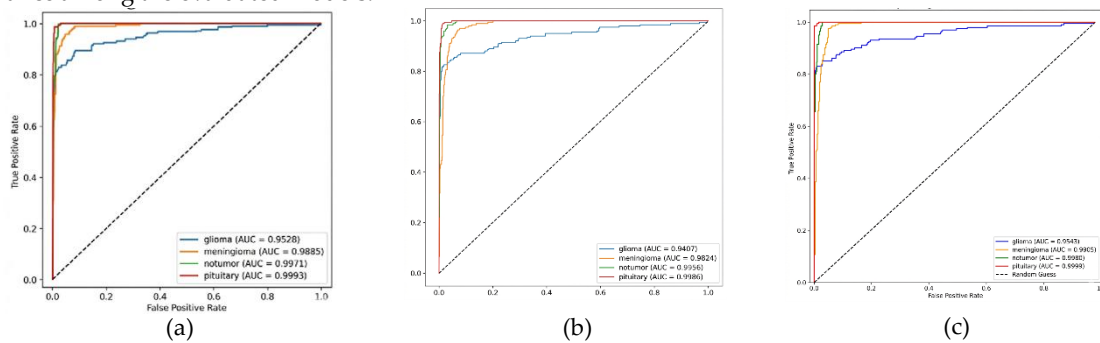


Figure 8 ROC curve of (a) CNN; (b) BCNN-MC Dropout; (c) BCNN-VI.

E. Uncertainty BCNN

To assess the reliability of the model's predictions, we quantify predictive uncertainty using two complementary metrics derived from T stochastic forward passes: predictive entropy, computed using Eq. (10), and predictive standard deviation, computed as the mean standard deviation of the class probabilities across the T passes. These two metrics together capture the model's *total* predictive uncertainty rather than a formal decomposition into aleatoric and epistemic components. Uncertainty was estimated separately for the MC Dropout and VI approximations, allowing for a direct comparison of how each Bayesian approximation behaves on the test set.

The distribution of predictive uncertainty differs noticeably between the two approximation methods. For the BCNN-MC Dropout model, the predictive entropy and standard deviation are generally higher and more widely spread across the test set, with an overall mean entropy of approximately 0.163. In contrast, the BCNN-VI model exhibits a markedly lower overall mean entropy (0.0903) and standard deviation (0.0137), with the majority of test samples concentrated at low uncertainty values, alongside a smaller cluster of samples exhibiting comparatively higher uncertainty as shown in Figure 9. This indicates that, on average, the VI approximation produces sharper and more confident predictive distributions than MC Dropout, although both models retain a subset of samples where uncertainty remains elevated,

consistent with prior findings showing that variational Bayesian methods yield better-calibrated and more reliable uncertainty estimates than MC Dropout-based approximation [20].

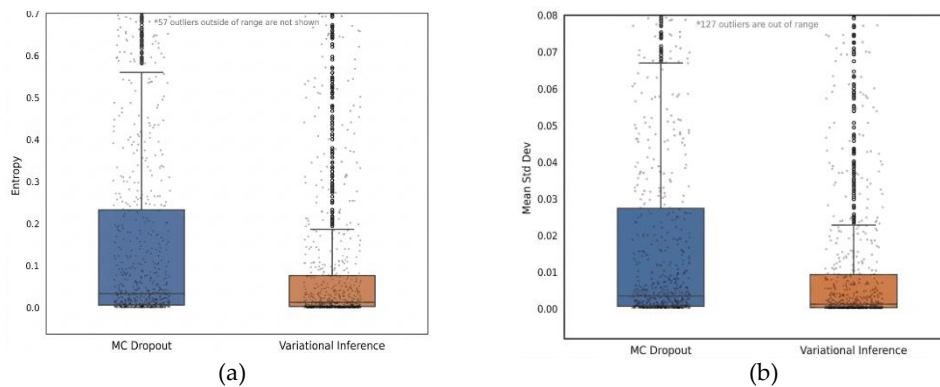


Figure 9 Boxplot predictive uncertainty MC-Dropout and VI of (a) Entropy; (b) Mean Standard Deviation.

Furthermore, Table 7 summarizes the average predictive entropy per tumor class for both Bayesian approximations. For the MC Dropout model, the meningioma class exhibits the highest predictive entropy (0.2787), suggesting that the model encounters the greatest ambiguity when distinguishing meningioma-related features from the other tumor types [19]. The Pituitary class shows the lowest entropy (0.0860), indicating the highest confidence among the four classes. For the VI model, the glioma class shows the highest entropy (0.1182), while the Pituitary class again exhibits the lowest entropy (0.0692). Notably, both approximation methods agree that the pituitary class is associated with the lowest predictive uncertainty, suggesting that this class possesses the most distinctive and easily separable features in the learned representation space, regardless of the Bayesian approximation method used. However, the two methods diverge in identifying the most uncertain class (meningioma for MC Dropout versus glioma for VI) indicating that each approximation captures a different notion of model ambiguity.

Table 7 Average Predictive Entropy per Tumor Class

Model	Glioma	Meningioma	No-tumor	Pituitary
BCNN-MC Dropout	0.1855	0.2787	0.1013	0.0860
BCNN-VI	0.1182	0.1035	0.0702	0.0692

To evaluate whether predictive uncertainty is a meaningful indicator of model reliability, we compare the average entropy and standard deviation between correctly and incorrectly classified test samples for both models, as summarized in Table 8. For both models, incorrectly classified samples exhibit substantially higher predictive entropy and standard deviation than correctly classified samples [12]. In the MC Dropout model, the mean entropy of incorrect predictions (0.5849) is approximately 4.7 times higher than that of correct predictions (0.1249), while in the VI model, incorrect predictions show a mean entropy (0.4160) approximately 5.8 times higher than correct predictions (0.0718). A similar pattern is observed for the predictive standard deviation in both models. This consistent gap demonstrates that both Bayesian approximation methods are able to meaningfully signal reduced confidence when the model is more likely to misclassify a sample, supporting the practical utility of predictive entropy and standard deviation as indicators of prediction reliability in this brain tumor classification task.

Table 8 Predictive Uncertainty for Correct vs Incorrect Predictions

Model	Mean Entropy (Correct)	Mean Entropy (Incorrect)	Mean Std (Correct)	Mean Std (Incorrect)
BCNN-MC Dropout	0.1249	0.5849	0.0163	0.0785
BCNN-VI	0.0718*	0.4160*	0.0108*	0.0648*

Overall, the two Bayesian approximation methods exhibit distinct uncertainty behaviors despite sharing the same underlying backbone architecture. The BCNN-VI model consistently produces lower and more concentrated uncertainty estimates across nearly all evaluation perspectives (overall distribution, per-class averages, and the correct/incorrect comparison), suggesting that the variational posterior over the network weights yields sharper and more confident predictive distributions than dropout-based sampling. Nevertheless, both methods show the same qualitative behavior in that higher uncertainty is reliably associated with misclassified samples, indicating that either approximation can serve

as a viable tool for flagging unreliable predictions in this classification task, with the choice between them involving a trade-off between the higher computational cost of VI (as established in Table 5) and its comparatively sharper uncertainty estimates.

Considering both classification performance and uncertainty quality, the BCNN-VI model is selected as the best-performing model in this study. Beyond exhibiting sharper and more concentrated uncertainty estimates, the VI model also achieves the highest test accuracy among the three models evaluated (95.00%), compared to 92.25% for the baseline CNN and 91.75% for BCNN-MC Dropout. This combination is particularly relevant in the context of medical image diagnosis, where classification accuracy is of primary importance, as misclassifying a tumor type can directly affect treatment decisions. At the same time, having access to well-concentrated and discriminative uncertainty estimates (rather than uncertainty values that are uniformly high and less informative) allows clinicians to better identify cases that warrant additional review or expert confirmation. The higher computational cost associated with training the VI model (Table 5) is therefore considered an acceptable trade-off, given its superior accuracy and more clinically interpretable uncertainty profile

F. BCNN-VI Predictive Uncertainty

To further illustrate how predictive uncertainty manifests at the individual sample level, Figure 10a and Figure 10b present the test samples with the highest and lowest predictive entropy under the BCNN-VI model, respectively. Each figure displays the input MRI image alongside the predictive class distribution, where the bar height represents the mean predicted probability for each class and the error bar represents the corresponding predictive standard deviation across the T stochastic forward passes.

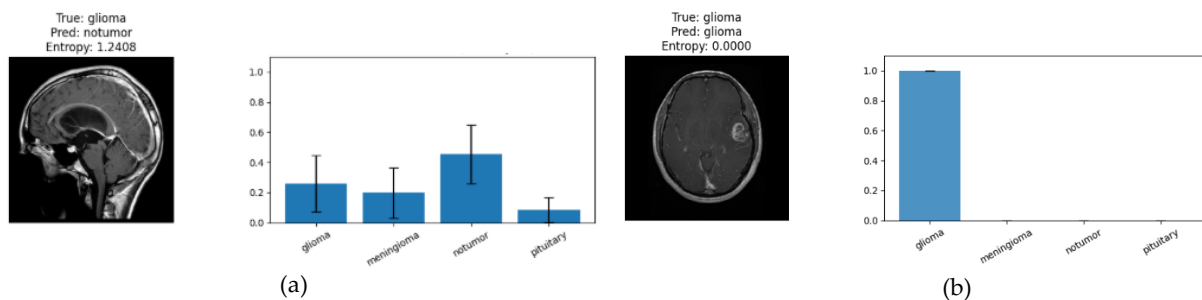


Figure 10 Individual sample predictive probability bars for BCNN-VI: (a) misclassified sample exhibiting split confidence, and (b) highly confident correct prediction

Figure 10a shows the test sample associated with the highest predictive entropy (entropy = **1.2408**), with a true label of **[Glioma]** and a predicted label of **[No-tumor]**. This sample was misclassified by the model, which is consistent with the earlier finding that higher predictive entropy is associated with a greater likelihood of misclassification (Table 8). The predictive distribution further reveals that the model's probability mass is spread across multiple classes rather than concentrated on a single class, accompanied by relatively large error bars, indicating substantial disagreement among the stochastic forward passes regarding the most likely class.

In contrast, Figure 10b presents the test sample with the lowest predictive entropy (entropy = **0.0000**), with a true label of **[Glioma]** and a predicted label of **[Glioma]**. This sample was correctly classified, and the predictive distribution shows a sharply concentrated probability mass on the correct class, with minimal probability assigned to the remaining classes and notably small error bars across all classes. This indicates strong agreement among the stochastic forward passes, reflecting high confidence in the model's prediction and suggesting that this image exhibits clear and easily distinguishable anatomical features characteristic of its tumor class.

These two illustrative cases demonstrate the practical interpretability of the predictive entropy as an uncertainty measure: low entropy corresponds to a sharply peaked, high-confidence predictive distribution typically associated with correct classification, whereas high entropy reflects a more diffuse distribution with greater disagreement across stochastic forward passes, often coinciding with classification errors or visually ambiguous tumor presentations. This sample-level evidence reinforces the dataset-level findings in Table 8, further supporting the use of predictive entropy as a practical indicator for flagging cases that may require additional clinical review.

V. CONCLUSIONS AND SUGGESTIONS

This study compared a deterministic baseline CNN with two Bayesian approximations, BCNN-MC Dropout and BCNN-VI, for classifying brain tumor MRI images into four classes. BCNN-VI achieved the best overall classification performance, with a test accuracy of 0.9463, outperforming the baseline CNN (0.9225) and BCNN-MC Dropout (0.9175), and demonstrated the most stable training and validation curves among the three models, indicating better

generalization despite requiring substantially more training time. Across all models, glioma remained the most frequently misclassified class due to overlapping imaging characteristics with meningioma and no-tumor, while pituitary and no-tumor were consistently classified with the highest accuracy.

Beyond classification accuracy, both Bayesian approximations successfully quantified predictive uncertainty through predictive entropy and standard deviation, with BCNN-VI producing sharper and more concentrated uncertainty estimates than BCNN-MC Dropout. Importantly, both methods showed that misclassified samples consistently exhibited substantially higher uncertainty than correctly classified ones, confirming that predictive uncertainty can reliably signal potentially unreliable predictions. This capability is clinically valuable, as it allows ambiguous or high-risk predictions to be automatically flagged for further review by a radiologist, rather than being accepted without scrutiny, positioning the Bayesian CNN not merely as a classifier, but as a decision-support tool that communicates its own confidence, which is essential for safe deployment in high-stakes diagnostic settings.

Future research may explore incorporating predictive uncertainty as a feedback signal within the network itself (for example, through uncertainty-aware loss weighting or uncertainty-guided curriculum learning) to determine whether leveraging uncertainty during training can further improve the model's predictive power rather than treating it solely as a post-hoc diagnostic measure. Additional directions include formally decomposing uncertainty into its aleatoric and epistemic components to better distinguish data-driven ambiguity from model-driven uncertainty, exploring hybrid architectures that apply variational layers selectively to reduce the computational cost of VI, and validating the proposed models on external, multi-institutional MRI datasets to assess their generalizability and clinical robustness beyond the single dataset used in this study.

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